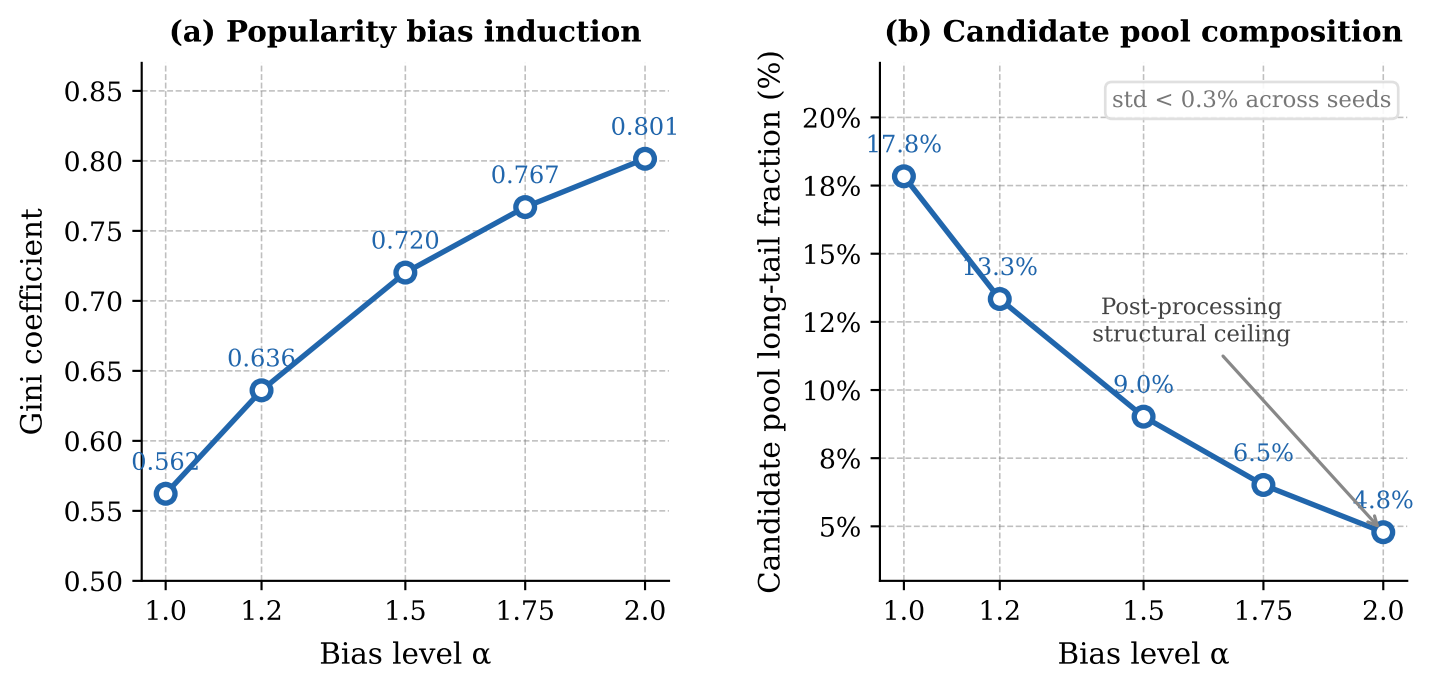
(a) Popularity bias induction
Gini coefficient
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.562
0.636
0.720
0.767
0.801
1.0
1.2
1.5
1.75
2.0
Bias level α
(b) Candidate pool composition
Candidate pool long-tail fraction (%)
std < 0.3% across seeds
20%
18%
15%
12%
10%
8%
5%
17.8%
13.3%
9.0%
6.5%
4.8%
Post-processing
structural ceiling
1.0
1.2
1.5
1.75
2.0
Bias level α